

Week 3 Seminar Tasks

Set Theory Basics

READING

Lecture 2 Notes (available on Blackboard)

Chapter 13. *Croft, T and Davison R (2016) Foundation maths, 6th ed. Harlow: Pearson.*

Chapter 5. *Grossman, P., 2017. Discrete mathematics for computing. Bloomsbury Publishing.*

TASK 1. INTRODUCTION TO SETS.

See Lecture 2 Notes – Slides 3-8.

Task 1.1. Set Construction. Given the following descriptions, construct the appropriate sets. Name the cardinality of these sets.

- The set of even natural numbers, which are less than 10.
- The set of vowels in the English alphabet.
- The set of months that have 31 days.

Task 1.2. Set Construction. Given the set $A = \{2,3,4,5,6,7,8,9\}$ form a new set B, which consists of all elements of set A that are:

- Prime numbers.
- Even numbers.
- IN-CLASS ACTIVITY** Odd numbers that are greater or equal to 3.
- IN-CLASS ACTIVITY** Those numbers that being multiplied by 2 give a number that is also an element of A.
- Those numbers that being multiplied by 2 give a number that is not in the given set A.
- Those numbers that being squared resulting in a number which also belongs to A.

Task 1.3. Set Cardinality

Given the sets:

- $F = \{\text{red, blue, yellow, }\{\}\}$
- $G = \{1, 2, 3, 4, 5, 6, 7\}$
- $H = \{A, B, C, D, E, F\}$
- $I = \{x, y, z, w\}$
- $J = \{\}$

- $K = \{\{H, B, \{M\}\}, \{I\}, \{J\}\}$

- a) **IN-CLASS ACTIVITY** What is $|F|$, $|G|$, $|H|$, $|I|$?
- b) Calculate $|G| - |H|$.
- c) Find $|I| + |J|$.
- d) Determine the cardinality of set K.

Task 1.4. Empty Sets Cardinality. Define the cardinality of the following sets:

1. $A = \{\{a, b\}, \{\}, \{c\}, \{d, e, \{f\}\}\}$
2. $B = \{1, 2, 3, \{\}, \{\}\}$
3. $C = \{\{\}, \{\}, \{\{\}, \{\}\}\}$

TASK 2. OPERATIONS WITH SETS

See Lecture 2 Notes – Slide 16-29

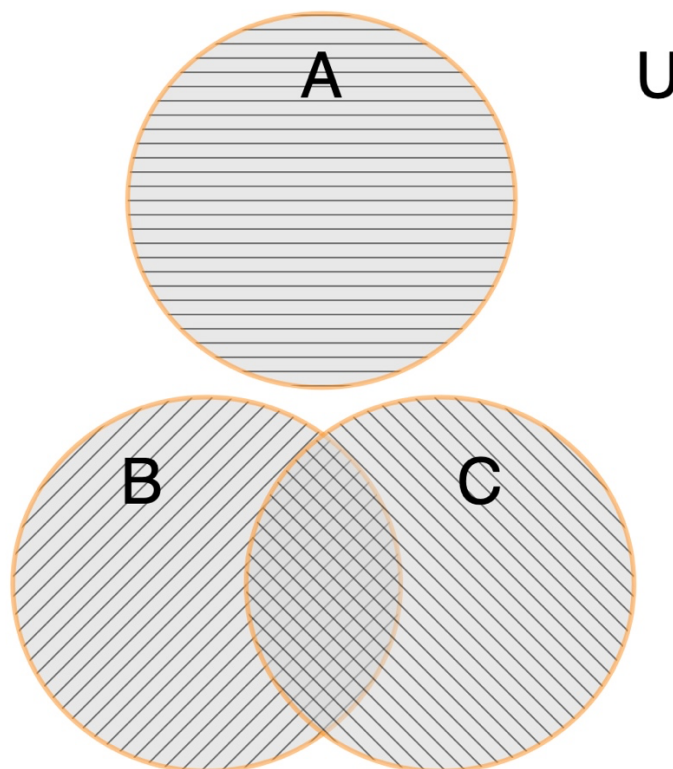
Task 2.1

IN-CLASS ACTIVITY Let A and B be the only sets in U and $A = \{5, 6, 10, 12\}$ and $B = \{5, 7, 11\}$. Apply the following operations to sets A and B

- a) $A \cup B$
- b) $A \cap B$
- c) $A \setminus B$
- d) $B \setminus A$
- e) A'
- f) B'
- g) $A' \cap B'$

Task 2.2

Consider sets $A = \{a\}$, $B = \{g, h, i, j\}$ and $C = \{i, j, k, l\}$. In the Venn diagram below, place the elements of the following sets and establish what sets result in the following operations:



- a) $B \cap C$
- b) $A \cup B$
- c) $A \cap B$
- d) $B \cup (B \cap C)$
- e) **IN-CLASS ACTIVITY** $A \cap (B \setminus U)$
- f) **IN-CLASS ACTIVITY** $U \cap (A \cup B)$

u means universal set so , all the sets are in the universal set , so with intersecting it still gives the same. $(A \cup B)$

Task 2.3

IN-CLASS ACTIVITY Below U is the universal set, $\{\}$ is the empty set and A is an arbitrary set. Based on the definition of the empty and universal sets, establish what should be the resulting set of the following operations:

- a) $A \cap U$
- b) $A \cup U$
- c) $A \setminus U$
- d) $(U \setminus A) \cup A$
- e) $\{\} \cup A$
- f) $\{\} \cap A$

Task 2.4

IN-CLASS ACTIVITY Let $A = \{a, b, c\}$ and $B = \{1, 0\}$.

- a) Write down all elements of the Power Set of A and Power Set of B . What is the Cardinality of $P(A)$ and $P(B)$?
- b) List all elements of $A \times B$
- c) List all elements of $B \times A$
- d) Calculate $|A \times B|$ and $|B \times A|$
- e) What is $A \times B \setminus B \times A$

TASK 3. SET OPERATIONS. APPLICATION IN COMPUTING

See Lecture 2 Notes – Slide 16-29

IN-CLASS ACTIVITY **Task 3.1: Organising Files.** Imagine you have two folders on your computer: Folder A contains [doc1, doc2, doc3, doc4], and Folder B contains [doc3, doc4, doc5, doc6]. Represent these files in sets and:

- Find the union and intersection of files in these two folders.
- Determine which files you would need to move to make the folders identical.

IDENTICAL MEANS EQUAL

Task 3.2 Managing Software Features. Imagine you are testing three versions of software: Version A with features [f2, f3, f6], Version B with features [f2, f3, f4], and Version C with features [f1, f5, f7].

- Identify which features are common across all versions and which are unique to each version.
- Find which features are present in at least two versions.

Task 3.3 Dataset Analysis. You are working as a data analyst at a multinational corporation that manages two e-commerce platforms: ShopA and ShopB. The company is interested in understanding the purchasing behaviours and preferences of its customers across these platforms to design targeted marketing campaigns. Your task is to perform a set theory-based analysis using the data provided in the tables below.

Table 1: Customers of ShopA

Customer ID	Purchased Electronics	Purchased Clothing	Purchased Groceries
1	Yes	No	Yes
2	No	Yes	No
3	Yes	Yes	Yes
4	No	Yes	Yes
5	Yes	No	No

Table 2: Customers of ShopB

Customer ID	Purchased Electronics	Purchased Clothing	Purchased Groceries
3	No	Yes	Yes
4	Yes	No	No
5	Yes	Yes	No
6	No	Yes	Yes

7	Yes	No	Yes
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- *Intersection Analysis:* Identify Customers who purchased "Electronics" in both ShopA and ShopB.
- *Union Analysis:* Identify Customers who purchased "Clothing" in either ShopA, ShopB, or both.
- *Difference Analysis:* Identify customers who purchased "Groceries" in ShopA but not in ShopB
- *Subset Analysis:* Identify if the set of customers who purchased all three items in ShopA is a subset of customers who purchased at least one item in ShopB. Explain your answer with set notations.

Note: The **universal set** comprises all customers in the corporation database, {1–7}.

TASK 4. SET BUILDER NOTATION. SUBSETS. SETS VS INTERVALS

See Lecture 2 Notes – Slide 9-15, 30-35

Task 4.1

In computing, we often have to analyse data of a given set (class) and transform data to a new set. This task simulates this scenario. Let N be a set of natural numbers $\{1,2,3,\dots\}$. For each case below, we define a new set B using set builder notation. Your task is to list all elements of B and to establish the cardinality of B .

- a) **IN-CLASS ACTIVITY** $B = \{x \mid x \in N \text{ and } x^2 = x\}$
- b) $B = \{x \mid x \in N \text{ and } x^2 = 2x\}$
- c) **IN-CLASS ACTIVITY** $B = \{(x,y) \mid x \in N \text{ and } y \in N \text{ and } x < y \text{ and } y \leq 3\}$
- d) $B = \{(x,y) \mid x \in N \text{ and } y \in N \text{ and } x = y \text{ and } y < 5\}$

Task 4.2

IN-CLASS ACTIVITY The following tasks are related to the concepts of interval, weak and strong subsets. Establish if the following statements are true or false.

- a) $3 \in (3, 5]$
- b) $5 \in (3, 5]$
- c) $\{1,2,3\} \subseteq \{3,2,1\}$
- d) $\{1,2,3\} \subset \{1,2,3\}$
- e) $[1,2] \subset \{1,2,3\}$
- f) $\{1,2,3\} \subset [1,2]$
- g) $\{-1,0,1\} \subseteq [-1,1]$

TASK 5. OPERATIONS ON SETS USING MODULAR ARITHMETIC

See Lecture 1 Notes for Modular Arithmetic revision

Let the sets A, B, and C be defined using modular arithmetic as follows:

$$A = \{x \mid x \equiv 2 \pmod{3}, 0 \leq x \leq 30\};$$

$$B = \{x \mid x \equiv 1 \pmod{4}, 0 \leq x \leq 30\};$$

$$C = \{x \mid x \equiv 0 \pmod{5}, 0 \leq x \leq 30\}.$$

1. **IN-CLASS ACTIVITY** Find all elements of sets A, B, and C by listing the numbers that meet the conditions in *the range from 0 to 30*.
2. Perform the following set of operations:
 - Find the intersection of A and B: $A \cap B$.
 - Find the union of A and C: $A \cup C$.
 - Find the difference between the union of B and C and set A: $(B \cup C) \setminus A$
3. Find the number of elements in each resulting set from part 2 (the cardinality).

TASK 6. INTERVALS

See Lecture 2 Notes – Slide 30-35

Task 6.1

Let $\mathbb{R}$ abbreviate a set of *real numbers*. Define the following intervals of real numbers using set builder notation:

- a) **IN-CLASS ACTIVITY** $[2,9]$
- b) $[2,9)$
- c) $(2,9]$
- d) $(2,9)$

Task 6.2

This task practices forming new intervals via set builder notation.

- a.) Let $A = [2,5]$ be a closed interval of all real numbers from 2 to 5. Which new interval is introduced by the following set builder notation: $\{x \in A: x < 4.5\}$
- b.) **IN-CLASS ACTIVITY** Let $A = [2,9]$ be a closed interval of all real numbers 2 to 9. Which new interval is introduced by the following set builder notation: $\{x \in A: \sqrt{x} \in A\}$

[CHALLENGES]

Challenge 1.1

Let $A = \{m \mid m \text{ is an integer satisfying } 0 < m < 13\}$ and $B = \{n \mid n \text{ is an integer satisfying } 7 < n < 23\}$.

Your task is to calculate $|(A \times B) \setminus (B \times A)|$.

Hint: if you start working on this problem, you will note that listing all the elements of the resulting set is not practical, so you need to figure out how to do it by applying some reasoning rather than listing ALL elements in the product sets.

Challenge 1.2

Let $A = [2, 9]$ be a closed interval of all real numbers 2 to 9. Which interval is introduced by the following set builder notation: $\{x \in A: \sqrt{x} \in A \text{ and } x < \sqrt{20}\}$?